6.1 Derivative function and usual functions

Definition and results

A function is detrievable over an interval when it is in each of its points; its derivative associates to x the number $f\prime(x)$ The usual derivatives are: constant $\mapsto 0$; $x \mapsto 1$; $x^2 \mapsto 2x$; $x^3 \mapsto 3x^2$; $\frac{1}{x} \mapsto -\frac{1}{x^2}$ for $x \neq 0$; $\sqrt{x} \mapsto \frac{1}{2\sqrt{x}}$ for $x > 0$. For $n \in \mathbb{Z}$, $(x^n)\prime = nx^{n-1}$ on its area of differentiability; n = 0, derivative is 0 directly.

Status and justification. Square: rate is 2 $a + h$. Inverse: for $a \neq 0$, $h \neq 0$ and $a + h \neq 0$, $\frac{[\frac{1}{a+h} - \frac{1}{a}]}{h} = -\frac{1}{[a(a+h)]}$, which tends to $-\frac{1}{a^2}$. The other formulas are allowed here.

Example solved. For $f(x) = x^{-3}$, the domain is $\mathbb{R} \setminus \{0\}$ and $f\prime(x) = -3x^{-4} = -\frac{3}{x^4}$. For $g(x) = \sqrt{x}$, the function exists in 0, its usual derivative only for $x > 0$.

Point of vigilance

The area of differentiability may be smaller than the area of definition.

Immediate application. Dirty x^5, x^{-2} and $\sqrt{x}$ specifying the areas.

6.2 Sum, product, inverse and quotient

Definition and results

For u and v derivables, ($u + v)\prime = u\prime + v\prime$, $(\lambda u)\prime = \lambda u\prime$, $(uv)\prime = u\prime v + uv\prime$. Where $v \neq 0$, $(\frac{1}{v})\prime = -\frac{v\prime}{v^2}$ and $(\frac{u}{v})\prime = \frac{u\prime v - uv\prime}{v^2}$.

Status and justification. For the product, add and cut $u(a + h)v(a)$ to the numerator. rate becomes $\frac{u(a+h)[v(a+h) - v(a)]}{h} + \frac{v(a)[u(a+h) - u(a)]}{h}$. As a drawable function is continuous (admitted), we get $u(a)v\prime(a) + v(a)u\prime(a)$ Other rules are allowed.

Example solved. For $f(x) = \frac{x^2+1}{x-1}$, $x \neq 1$, $f\prime(x) = \frac{[2x(x-1) - (x^2+1)]}{(x-1)^2} = \frac{x^2 - 2x - 1}{(x-1)^2}$. Keeping parentheses avoids changing a single sign instead of the entire numerator.

Point of vigilance

The derivative of a product is not the product of the derivatives; the derivative of a quotient is not their quotient.

Immediate application. Derivative $(x^2+1)(3x-2)$ then $\dfrac{1}{x^2+1}$.

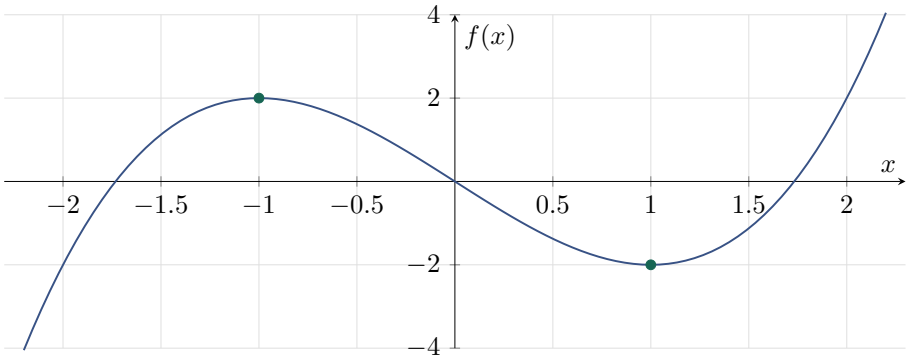
6.3 Sign of derivative and extrema

Definition and results

In an interval I, if $f' \geq 0$, f is increasing; if $f' \leq 0$, is decreasing; if $f' = 0$ everywhere, it is constant. A strictly positive derivative implies strict increase. If f has a local extremum at an inner point where it is detrievable, its derivative is zero. Reciprocality is false.

Status and justification. Theorems of variation allowed. For an interior minimum of a, rates on the right are positive or null, those on the left are negative or null; if they converge towards the same real, it Same argument for a maximum.

Example solved. $f(x) = x^3 - 3x$: $f'(x) = 3(x-1)(x+1)$. f grows on $]-\infty; -1]$, decreases on $[-1; 1]$, grows on $[1; +\infty[$; local maximum $f(-1) = 2$, local minimum $f(1) = -2$. None of these extrema n is global $\mathbb{R}$.



$f(x) = x^3 - 3x$: a local maximum in -1 and a local minimum in 1.

Point of vigilance

The theorem applies over an interval. A function may decrease over two intervals separated without decreasing on their meeting.

Immediate application. Study $f(x) = x^2 - 4x + 1$ on $[0; 3]$ and give his extrema on this Interval.

6.4 Parity, Inequalities and Comparison

Definition and results

On a symmetric domain relative to 0, f is pairs if $f(-x) = f(x)$ and odd if $f(-x) = -f(x$The curve of a pair function is symmetrical in relation to the axis of the
In order to compare f and g, the sign is studied
of $f - g$, possibly with its derivative.

Status and justification. The symmetries are derived from the coordinates of the image points. Inequality, a study of variations can give a minimum of the difference, then its sign; the derivative does not substitute directly for this difference.

Example solved. For $x > 0$, $h(x) = x + \frac{1}{x} - 2$a for derivative $\frac{x^2-1}{x^2}$. It decreases up to 1 then grows; $h(1) = 0$, therefore $x + \frac{1}{x} \geq 2$. Equality is achieved only in 1.

Point of vigilance

Verify the domain before parity. $f\prime - g\prime$ information on the change in $f - g$, not directly on his sign.

Immediate application. Studying gender parity $x^4 + x^2$ and $x^3 - x$; Demonstrate $x^2 + 1 \geq 2x$.

6.5 Newton Optimization and Method

Definition and results

Optimize is to define the variable and its domain, express the size to optimize, then compare all the candidate values, including the bounds.

uses tangent to approach a solution of $f(x) = 0 : x_{n+1} = x_n - \frac{f(x_n)}{f}\prime(x_n)$ where $f\prime(x_n) \neq 0$. It is used only in favourable cases.

Status and justification. The Newton formula comes from the intersection with the horizontal axis of $y = f(x_n) + f\prime(x_n)(x - x_n)$ Convergence is not guaranteed in general; it is recognized and monitored in the case chosen.

Example solved. For $f(x) = x^2 - 2$ with $x_0 = 1$, Newton gives $x_1 = \frac{3}{2}$, $x_2 = \frac{17}{12}$. Squares are 2.25 then about 2,00694 : we are approaching $\sqrt{2}$. The program must control derivative zero and number maximum steps.

Point of vigilance

A numerical value provided by an algorithm is not proof of minimum or convergence.

Immediate application. A perimeter rectangle 12 m wide x. Determine maximum area by diversion.